

Inspiring Excellence

CSE421 : Computer Networks
Project Report
Group No: 1301
Sec#13

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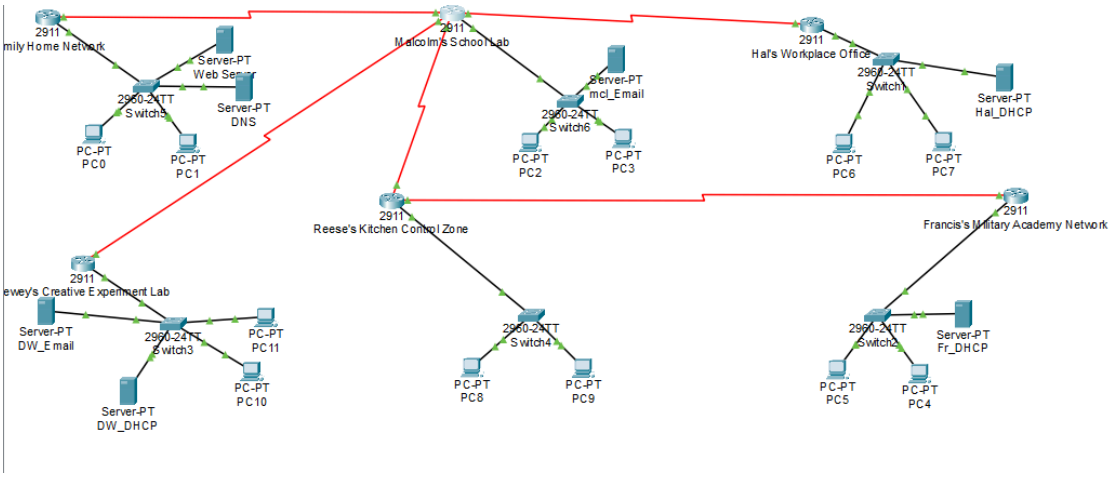
Tasneem Tuhfa

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Introduction

The following report discusses a network topology created for Malcolm and his family to set up communication across home, school, workplace and other important locations.

1. Network Topology



2.Router Configuration Commands

A. Malcolm-R

```
en
config t
hostname Malcolm-R
!
ip dhcp excluded-address 22.29.0.1 22.29.0.10
!
ip dhcp pool MALCOLM_LAN
 network 22.29.0.0 255.255.252.0
 default-router 22.29.0.1
 dns-server 22.29.9.131
!
interface GigabitEthernet0/1
 ip address 22.29.0.1 255.255.252.0
 no shutdown
```

```
!
interface Serial0/0/0
 ip address 22.29.9.162 255.255.255.252
 clock rate 2000000
 no shutdown
!
interface Serial0/0/1
 ip address 22.29.9.170 255.255.255.252
 clock rate 2000000
 no shutdown
!
interface Serial0/1/0
 ip address 22.29.9.166 255.255.255.252
 no shutdown
!
interface Serial0/1/1
 ip address 22.29.9.174 255.255.255.252
 no shutdown
!
router rip
 version 2
 redistribute static
 network 22.0.0.0
 no auto-summary
!
ip route 22.29.4.0 255.255.254.0 22.29.9.173
ip route 22.29.8.0 255.255.255.0 22.29.9.173
ip route 22.29.9.0 255.255.255.128 22.29.9.169
ip route 22.29.9.128 255.255.255.224 22.29.9.161
ip route 22.29.6.0 255.255.254.0 22.29.9.165
!
```

B. Francis-R

```
en
config t
hostname Francis-R
!
interface GigabitEthernet0/1
 ip address 22.29.4.1 255.255.254.0
 no shutdown
!
interface Serial0/0/0
 ip address 22.29.9.177 255.255.255.252
```

```
 no shutdown
!
ip route 22.29.9.128 255.255.255.224 22.29.9.178
ip route 0.0.0.0 0.0.0.0 22.29.9.178
!
```

C. Hal-R

```
en
config t
hostname Hal-R
!
interface GigabitEthernet0/0
 ip address 22.29.9.1 255.255.255.128
 no shutdown
!
interface Serial0/0/0
 ip address 22.29.9.169 255.255.255.252
 no shutdown
!
router rip
 version 2
 network 22.0.0.0
 no auto-summary
!
```

D. Reese-R

```
en
config t
hostname Reese-R
!
ip dhcp excluded-address 22.29.8.1 22.29.8.10
!
ip dhcp pool REESE_LAN
 network 22.29.8.0 255.255.255.0
 default-router 22.29.8.1
 dns-server 22.29.9.131
!
interface GigabitEthernet0/1
 ip address 22.29.8.1 255.255.255.0
 no shutdown
!
interface Serial0/0/0
```

```
 ip address 22.29.9.178 255.255.255.252
 clock rate 2000000
 no shutdown
!
interface Serial0/0/1
 ip address 22.29.9.173 255.255.255.252
 clock rate 2000000
 no shutdown
!
router rip
 version 2
 redistribute static
 network 22.0.0.0
 no auto-summary
!
ip route 22.29.4.0 255.255.254.0 22.29.9.177
ip route 22.29.9.128 255.255.255.224 22.29.9.174
ip route 22.29.0.0 255.255.252.0 22.29.9.174
ip route 22.29.9.160 255.255.255.252 22.29.9.174
ip route 22.29.9.164 255.255.255.252 22.29.9.174
ip route 22.29.9.168 255.255.255.252 22.29.9.174
!
```

E. Dewey-R

```
en
config t
hostname Dewey-R
!
interface GigabitEthernet0/1
 ip address 22.29.6.1 255.255.254.0
 no shutdown
!
interface Serial0/0/0
 ip address 22.29.9.165 255.255.255.252
 clock rate 2000000
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 22.29.9.166
!
```

F. Family_R

```
en
```

```
config t
hostname Family_R
!
interface GigabitEthernet0/0
 ip address 22.29.9.129 255.255.255.224
 no shutdown
!
interface Serial0/0/0
 ip address 22.29.9.161 255.255.255.252
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 22.29.9.162
ip route 22.29.4.0 255.255.254.0 22.29.9.162
ip route 22.29.6.0 255.255.254.0 22.29.9.162 25
ip route 22.29.8.0 255.255.255.0 22.29.9.162
ip route 22.29.9.0 255.255.255.128 22.29.9.162
!
exit
```

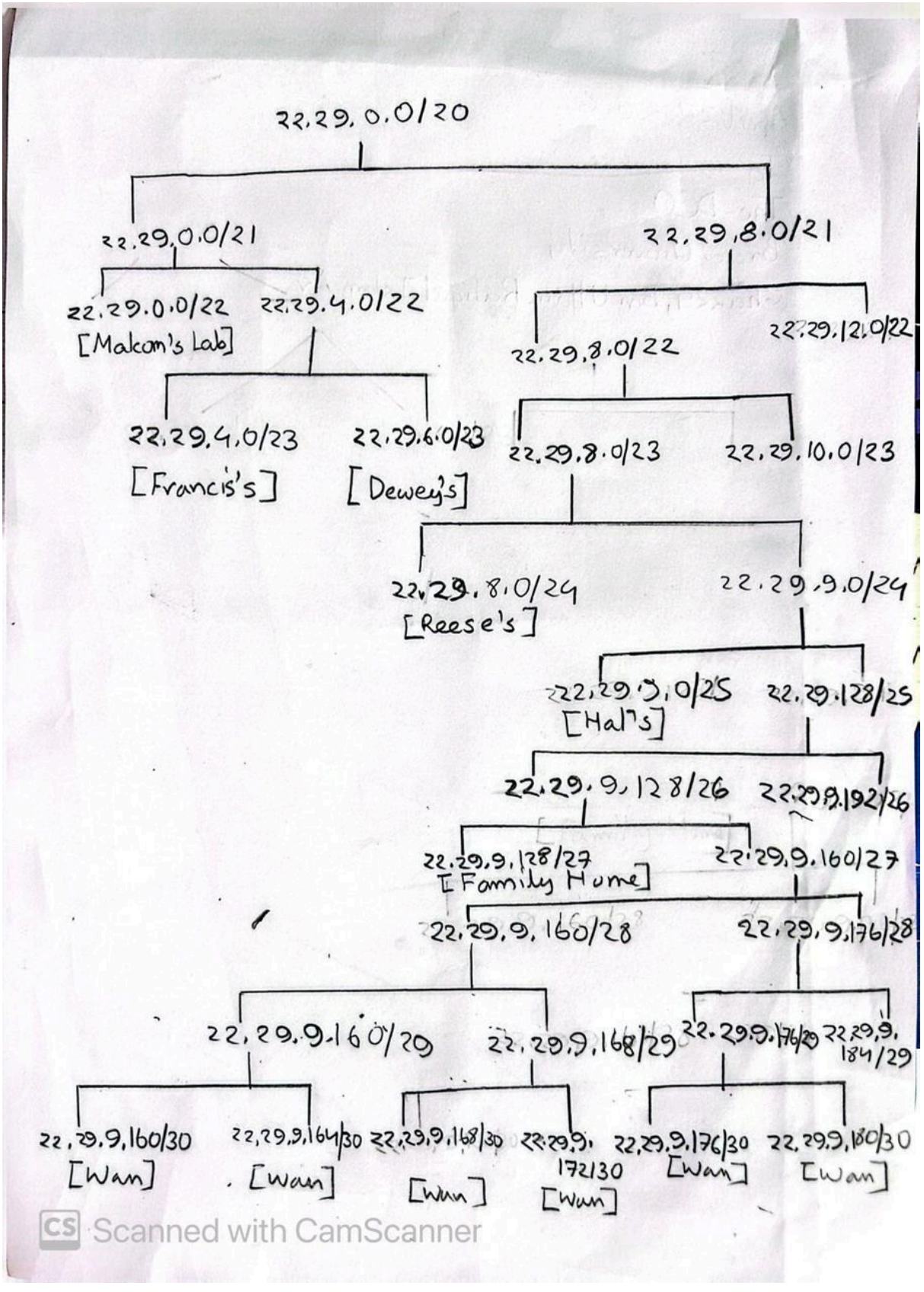
G. Verification Commands

Show ip route

```
Gateway of last resort is not set

     22.0.0.0/8 is variably subnetted, 16 subnets, 7 masks
S       22.29.4.0/23 [1/0] via 22.29.9.173
S       22.29.6.0/23 [1/0] via 22.29.9.165
S       22.29.8.0/24 [1/0] via 22.29.9.173
S       22.29.9.0/25 [1/0] via 22.29.9.169
S       22.29.9.128/27 [1/0] via 22.29.9.161
R       22.29.9.176/30 [120/1] via 22.29.9.173, 00:00:11, Serial0/1/1
```

3. VLSM Tree



Scanned with CamScanner

4. IP Address Table

Student Id: 22299074 . We can break our student ID down to form the first two octets of our starting IP address.

- Taking the first two numbers: 22
- Taking the next two numbers: 29

This gives us a starting base network address of `22.29.0.0`.

the first usable IP is typically assigned to the Router Interface for that network :

Network Name	Hosts Needed	Subnet Mask (CIDR)	Network Address	First Usable IP (Gateway)	Last Usable IP	Broadcast Address
1. Malcolm's School	840	255.255.252.0 /22	22.29.0.0	22.29.0.1	22.29.3.254	22.29.3.255
2. Francis's Academy	390	255.255.254.0 /23	22.29.4.0	22.29.4.1	22.29.5.254	22.29.5.255
3. Dewey's Lab	300	255.255.254.0 /23	22.29.6.0	22.29.6.1	22.29.7.254	22.29.7.255
4. Reese's Kitchen	210	255.255.255.0 /24	22.29.8.0	22.29.8.1	22.29.8.254	22.29.8.255

5. Hal's Workplace	72	255.255.255.128/25	22.29.9.0	22.29.9.1	22.29.9.126	22.29.9.127
6. Family Home	25	255.255.255.224/27	22.29.9.128	22.29.9.129	22.29.9.158	22.29.9.159

The WAN Links (Router-to-Router) :

WAN Link Connection	Mask	Network Address	IP Address 1 (Router A)	IP Address 2 (Router B)
WAN 1 (Home to Malcolm)	/30	22.29.9.160	22.29.9.161	22.29.9.162
WAN 2 (Dewey to Malcolm)	/30	22.29.9.164	22.29.9.165	22.29.9.166
WAN 3 (Hal to Malcolm)	/30	22.29.9.168	22.29.9.169	22.29.9.170
WAN 4 (Reese to Malcolm)	/30	22.29.9.172	22.29.9.173	22.29.9.174

WAN 5 (Francis to Reese)	/30	22.29.9.176	22.29.9.177	22.29.9.178

5. Derivation of Base Network Address

The base network address for this was derived from the student ID 22299074 where the first two digits of the id were used as the first octet and the next two id digits were used as the second octet of the base network address. The rest of the octets are kept as 0.

6. Brief Notes

DHCP Configuration

Router-Based DHCP:

- **Malcolm's School Lab** — DHCP configured directly on the router interface serving that subnet
- **Reese's Kitchen Control Zone** — DHCP configured directly on the router interface serving that subnet

Server-Based DHCP:

- **Francis's Military Academy Network** — a dedicated DHCP server within the subnet distributes addresses
- **Hal's Workplace Office** — a dedicated DHCP server within the subnet distributes addresses
- **Dewey's Creative Experiment Lab** — a dedicated DHCP server within the subnet distributes addresses

Routing

Routers using Static Routing:

- **Family Home Router**
- **Francis's Military Academy Router** — configured with standard static routes pointing back to the Family Home Network

Routers using Dynamic Routing :

- Malcolm's School Lab
- Reese's Kitchen Control Zone
- Hal's Workplace Office

Hybrid Router (Both Static and Dynamic Routing)

- **Malcolm's School Lab Router** is the hybrid router. It runs the dynamic routing protocol (RIP) with Reese's Kitchen Zone and Hal's Workplace, and simultaneously handles static routing — receiving the default static route from the Family Home Router and serving as the next-hop for the floating static route to Dewey's Lab. This makes it the central convergence point of the entire topology where both routing paradigms coexist on the same device.